

1. Assessment Style

- **Homework** – Four questions given every three weeks in advance of lecture material. Homework will be hosted on our tutorial's Postal Owl page. submissions will only be accepted through OWL as either Markdown with KaTeX math or PDF.

Submissions may be repeated even after an initial grade is awarded, but points will not be awarded to solutions submitted after the due date unless explicit permission is given *well in advance*.

- **Oral Presentation** – During the tutorial session following the due date for a homework, students will be asked to group up, formulate a collaborative solution and present it to the rest of the class. An exam-style grade will be given: 1 *excellent*, 2 *very good*, 3 *pass* and 4 *fail*.

Given our presentation structure and strict due dates, I strive to respond to homework submissions as quickly as possible. **Owl will be checked at 10:00 am every Monday, Tuesday, Thursday and Friday**; submissions made by that time will be graded and responded to within the day. **Please use this availability to ask questions early and often.** Contact Jamie if you do not yet have access to our Owl page.

2. Credit Requirement

- Earn at least 50% of the available homework points.
- Present one problem from each homework.
- Score a 4 (failure) on no more than one of said presentations.

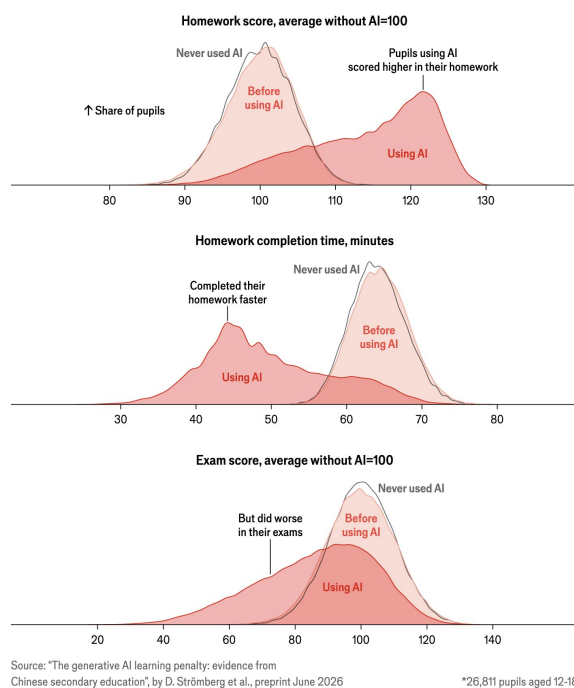
Any student who feels that they may need accommodation because of a disability (learning disability, attention deficit disorder, social anxiety disorder, etc.), please contact me as soon as possible.

3. Collaboration is Encouraged

On any assignment, you are encouraged to work with other students and to freely ask questions of the instructor online or in person. If your goal is to learn or to do well on the exam, I think it is better for you to talk with real people—other students or a human teacher—rather than to any sort of artificial intelligence.

4. Artificial Intelligence is Not

Recent research says: if you are outsourcing your homework to an AI, then you are learning less, you aren't improving your endurance, and you will struggle on the exam. I cannot monitor AI usage, but I am trying very hard to make himself available to you; please take advantage of that!



More than half of AI students spend less time than even the fastest non-AI students. This group receives very high homework scores [and] extremely low exam scores.

After six months of use, homework scores rise by 18 percent of the baseline mean while completion time per assignment falls from 64 to 45 minutes. However, students' scores in monthly school exams fall by 20 percent of the baseline mean.

Our finding that exam scores deteriorate gradually over time suggests that homework outsourcing may undermine not only the retention of current material but also the absorption of future material [2]. ... It is also consistent with recent evidence on "cognitive endurance," the capacity to sustain effortful attention in improving cognitive ability [1].

Stromberg, Lei, and Wu [3]

Algorithms and Data Structures II

Tutorial Syllabus (continued)

5. Homework Suggestions Clarity is important and will often require introducing some notation of your own. Assign precise names (A or B , x or y) to objects rather than calling them by pronouns ("this set" or "that vector"). You are strongly encouraged to learn the basics of writing your math in as it is the standard typesetting language in every mathematical field. OWL supports both Markdown and KaTeX inputs; documentation on both are linked above the editing box of the homework assignments!

<https://wordpress.com/support/markdown-quick-reference/>

<https://katex.org/docs/supported.html>

Genuine collaboration is one of the most effective ways to learn anything. I strongly encourage collaborating with peers when solving homework tasks, but genuine collaboration requires that everyone is honest and contributes to the conversation. Here are some general guidelines:

- Try to solve the task on your own before collaborating. You cannot really contribute if you haven't tried it.
- Give other students space to talk. They will have insights that you did not see.
- *Always submit your own solutions!* Duplicate submissions will be reported and treated as plagiarism.

6. Course Guidelines

- Attendance is not required and will not contribute to your grade beyond the need to give presentations in person.
- Plagiarism is strictly forbidden and will be punished with course failure and possibly with university discipline. In this instance, plagiarism includes duplicate submissions between students and AI generated submissions.
- Be respectful of your fellow students. Bullying, jeering or other disrespectful behavior will see you removed from the classroom and potentially earning a failing grade.

7. Tentative Calendar

Week		Topic		Presentations	
1	05/10	String Search	KMP		
2	12/10	String Search	AC		
3	19/10	String Search	RK		
4	26/10	Network Flows	FF	HW1: String Search	
5	02/11	Network Flows	Dinic		
6	09/11	Network Flows	PR1		
7	16/11	Fast and Parallel	FFT	HW2: Network Flows	*Tuesday Tutorial Canceled
8	23/11	Fast and Parallel	Addit		
9	30/11	Fast and Parallel	PSort		
10	07/12	Geo. Algorithms	CHull	HW3: Fast and Parallel	
11	14/12	Geo. Algorithms	PlSwp		
12	04/01	Hardness			
	10/01			HW4: Geo and Hardness	*End of Semester: 08/01 *Start of Exams: 11/01

References

- [1] Christina Brown et al. "Cognitive endurance as human capital". In: *The Quarterly Journal of Economics* 140.2 (2025), pp. 943–1002. DOI: [10.1093/qje/qjae043](https://doi.org/10.1093/qje/qjae043).
- [2] James J Heckman and Chase O Corbin. "Capabilities and skills". In: *Journal of human development and capabilities* 17.3 (2016), pp. 342–359. DOI: [Capabilitiesandskills](https://doi.org/10.1016/j.jhealecap.2016.03.001).
- [3] David Stromberg, Victor Lei, and Yanhui Wu. "The Generative AI Learning Penalty: Evidence from Chinese Secondary Education". In: *SIAM Journal on Computing* (2026). DOI: [10.2139/ssrn.6868618](https://doi.org/10.2139/ssrn.6868618).

1. String Rotation A *rotation* of a string α is the string $\alpha[k :]\alpha[: k]$ (the last $|\alpha| - k$ characters of α preceding its last k) for some index k .

- Propose a linear-time algorithm which, given two strings α and β , decides whether α is a rotation of β for some k .
 - It should be relatively clear that a **True** response implies that α is a rotation of β . Prove the following:
Proposition. *If my process returns False, then α is not a rotation of β .*
 - Why does your process fail when looking for a rotation of β contained in α ? Give an example.
- ★★ How long might it take to find a rotation of β contained in α ?

- The time complexity of *my* solution is $\mathcal{O}(\min(|\alpha|, |\beta|))$.
- Contraposition ($A \rightarrow B \equiv \neg B \rightarrow \neg A$)

- Verify that $|\alpha| = |\beta|$ before using the Knuth-Morris-Pratt algorithm to check for β as a substring of the concatenation $\alpha\alpha$. Verifying the sizes can be done in $\mathcal{O}(\min(|\alpha|, |\beta|))$ by iterating through both strings until one ends, then KMP will run in the same time complexity. The value k will be the index corresponding to the first entry of the substring.
 - Proof.* The contrapositive “If α is a rotation of β , then my process returns **True**” is logically equivalent to the proposition statement.

Notice that, if α is indeed a rotation of β , then $\alpha\alpha = \beta[k :]\beta[: k]\beta[k :]\beta[: k] = \beta[k :]\beta\beta[: k]$ must contain β as a substring regardless of the value of k . KMP would find this substring and return **True**. $\square$
 - If $\alpha = \mathbf{f}\beta[k :]\beta[: k]$, then $\alpha\alpha = \mathbf{f}\beta[k :]\beta[: k]\mathbf{f}\beta[k :]\beta[: k]$ no longer contains β despite α containing a rotated β .
- ★★ A **False** instance requires searching for each of the $|\beta|$ rotations of β in α and necessarily takes quadratic time.

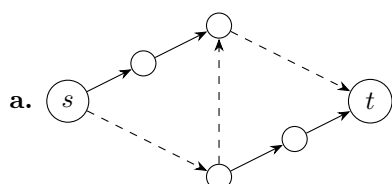
2. Maximum Disjoint s - t Paths Given a directed network with source s and sink t , we want to find a maximum collection of edge-disjoint, $s - t$ paths. Consider the following procedure:

- Run BFS rooted at s until t is reached for the first time.
- Identify the shortest $s - t$ path p by following parents back to s .
- Remove p from the graph and add it to P .
- Repeat until s and t lie in separate components.

The set P is clearly a collection of edge-disjoint, $s - t$ paths since no edge can be added to P more than once.

- Provide an example network on which this procedure fails to produce a *maximum* solution.
 - Propose a revised procedure which produces the correct set on any network and prove the following:
Proposition. *The collection produced by the revised procedure is maximum.*
 - What if the paths have to be internally vertex-disjoint (except for s and t)?
- ★★ What needs to change for use on an undirected graph?

- Eventually, removing the shortest path disconnects the graph. Can this happen too early?
- Reduce P to a network flow. Proof by contradiction?



Breadth-first search always finds the shortest path from s to t . In this case, s and t are disconnected after only one path is identified despite two edge-disjoint paths being possible.

- Assign unit capacity to each edge, find a maximum $s - t$ flow and remove unsaturated edges before performing the original procedure. Notice that the central, vertical edge above will be removed by this pre-processing step and the correct, two-path solution will be found.

Proof. Clearly, any collection P can be converted into a unit-flow by saturating those edges which

Example Problems (continued)

appear anywhere in P ; this flow will have value equal to the number of paths in P . It remains only to show that the collection P^* produced by the revised procedure translates into a maximum unit-flow.

Suppose, for the sake of contradiction, that P^* translates into a sub-optimal unit-flow. Since the flow F^* produced during the pre-processing step is necessarily maximum, this implies that there exists an edge e which is saturated by F^* but is not used by any path in P^* . Because both flow and elements of P^* must originate at s and end in t , we have a contradiction. Either e must be part of an $s - t$ path which was never added to P^* or the flow through e is imbalanced. $\square$

- c. Add unit capacity to each vertex. Ford-Fulkerson can still be used if we split each vertex into two vertices joined by a unit capacity edge.
- ★★ Treat undirected edges as two entangled edges in opposing directions; both have a capacity of one, but saturating one adds capacity to the other. Keep only those saturated directed edges for the second stage.